CS 4782 · DEEP LEARNING · FINAL POSTER

# Multi-Condition Diffusion for Paired Antibody Sequences

*A decoding sweep reveals a quality–diversity Pareto frontier*

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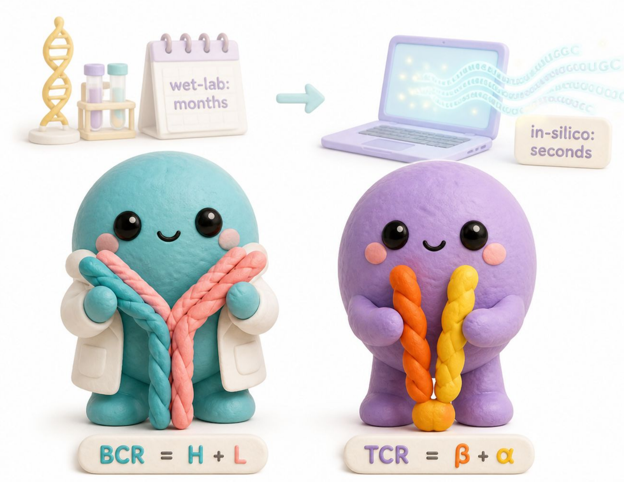
## §1 Background and Motivation

### Why BCR / TCR prediction?

- Paired heavy + light chain antibodies (VH+VL) target pathogens through dual-chain recognition
- Wet-lab discovery is slow and expensive (~$2.6B and 10+ years per drug)

### Why diffusion?

✓ Iterative refinement with global context at every step
✓ Native conditional control via classifier-free guidance





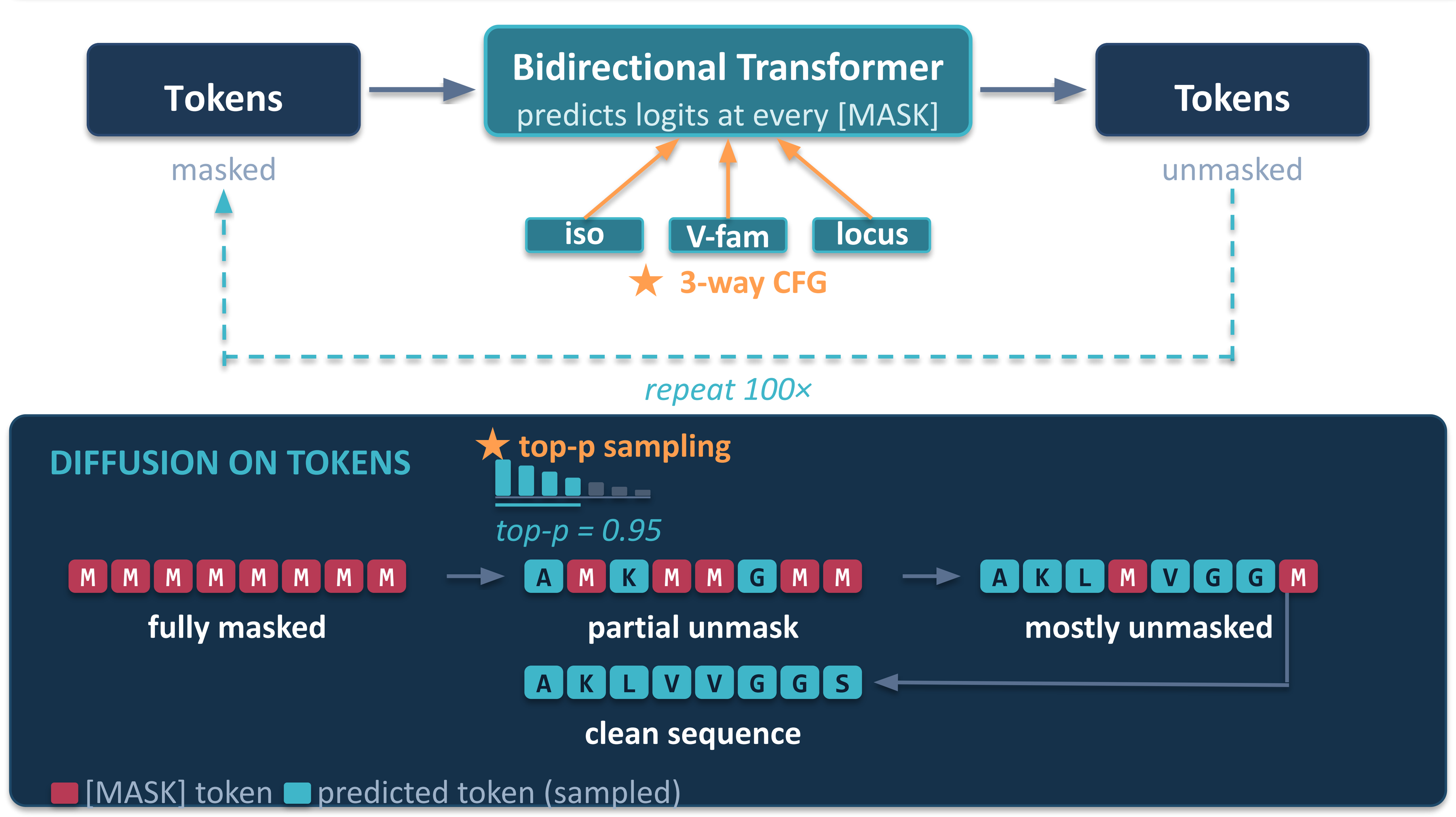
## §2 Antibody Dataset

**★ Dataset Not From Paper**

- 2.17M paired human antibody sequences(BCR)
- **MMseqs2 similarity reduction** (95% / 90%) + 0% leakage

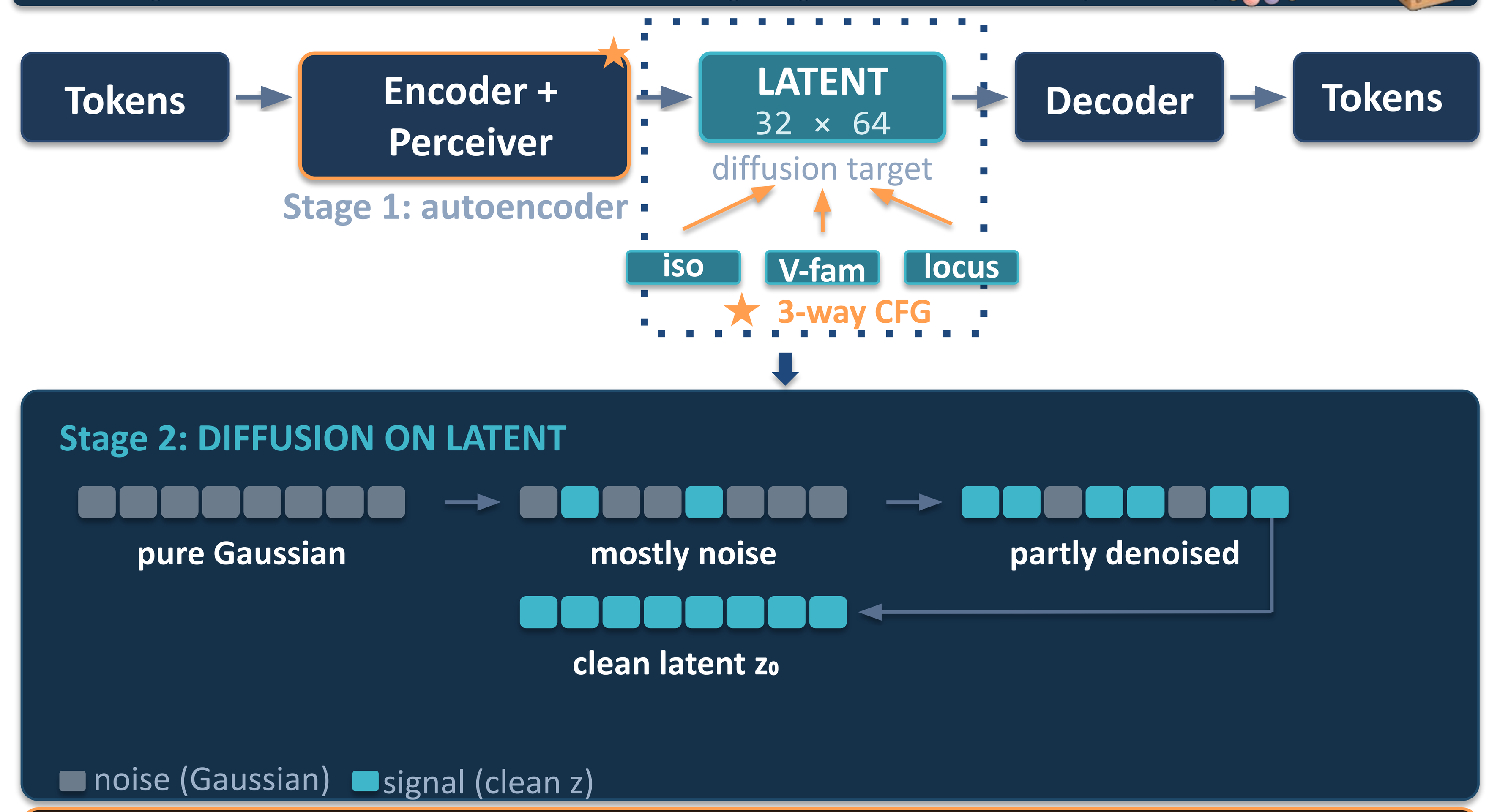
## §3 Methodology - Two diffusion paradigms

**Paradigm 1 : Diffusion Protein Language Model (DPLM)**




**★ EXTENSIONS BEYOND PAPER**
① 3-way CFG ② stochastic top-p decoder ③ 20-config T × top-p sweep

**Paradigm 2 : Latent Diffusion for Language Generation (LD4LG)**



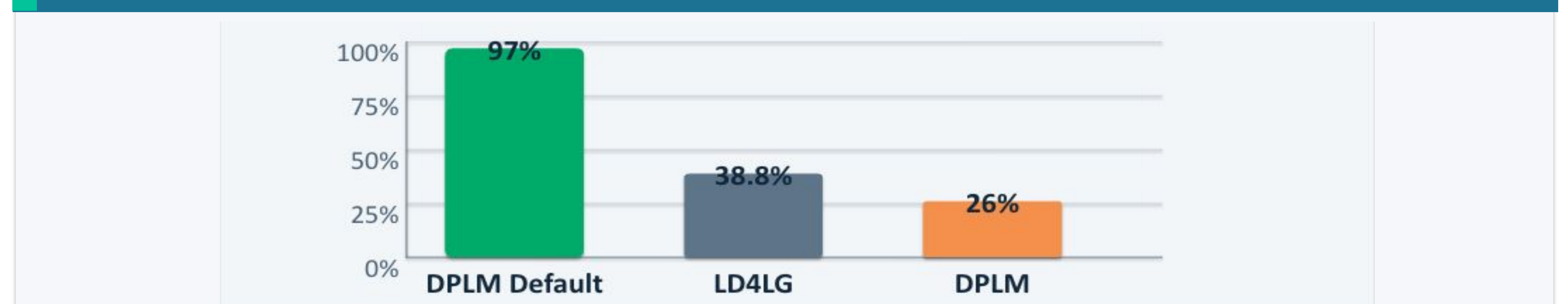

**★ EXTENSIONS BEYOND PAPER**
① AE trained from scratch ② 3-way CFG ③ self-cond disabled (DDP)

## §4.1 Results: Quality–Diversity Trade-off

| Metric *(Bio/Actual Term)* | DPLM (default · T=1.0, p=0.95) | ★ DPLM + sweep (tuned · T=1.3, p=0.99) | LD4LG |
|---|---|---|---|
| Format correctness *(linker recovery)* | 100.0% | 92.7% | 99.7% |
| Sequence diversity *(4-gram diversity)* | 0.051 | **0.207** | 0.136 |
| Conditional accuracy *(V-family class)* | **99.98%** | 93.8% | 99.6% |
| Sequence recovery rate *(SRR)* | 0.476 | 0.564 | **0.753** |
| Model perplexity *(held-out NLL)* | 1.37 | 1.37 | **1.08** |
| Domain validity *(HMMER match rate)* | 100% | 99.5% | 100% |

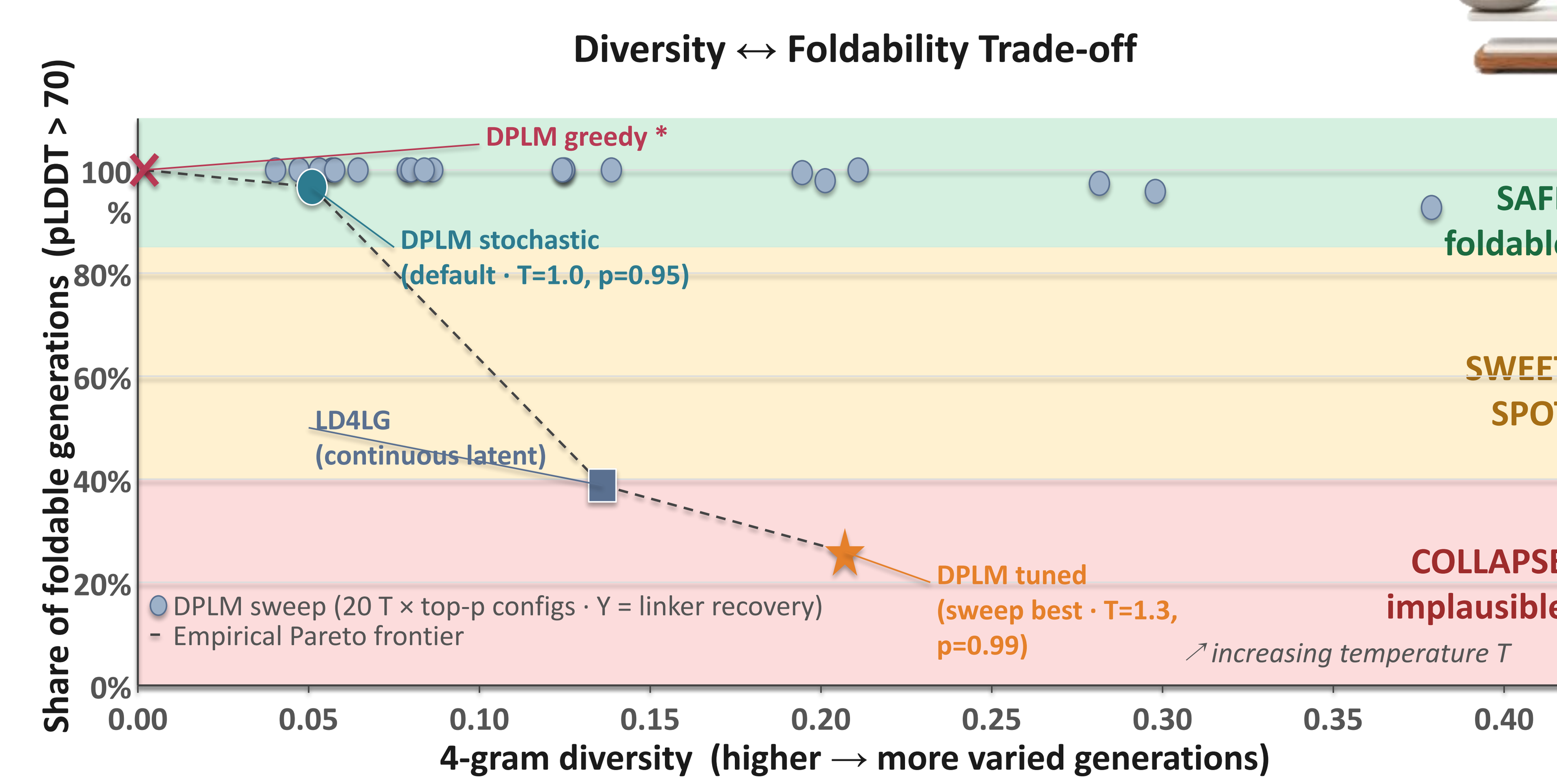
*Best Score in Bold

## §4.2 Results: Foldable Share Across Models

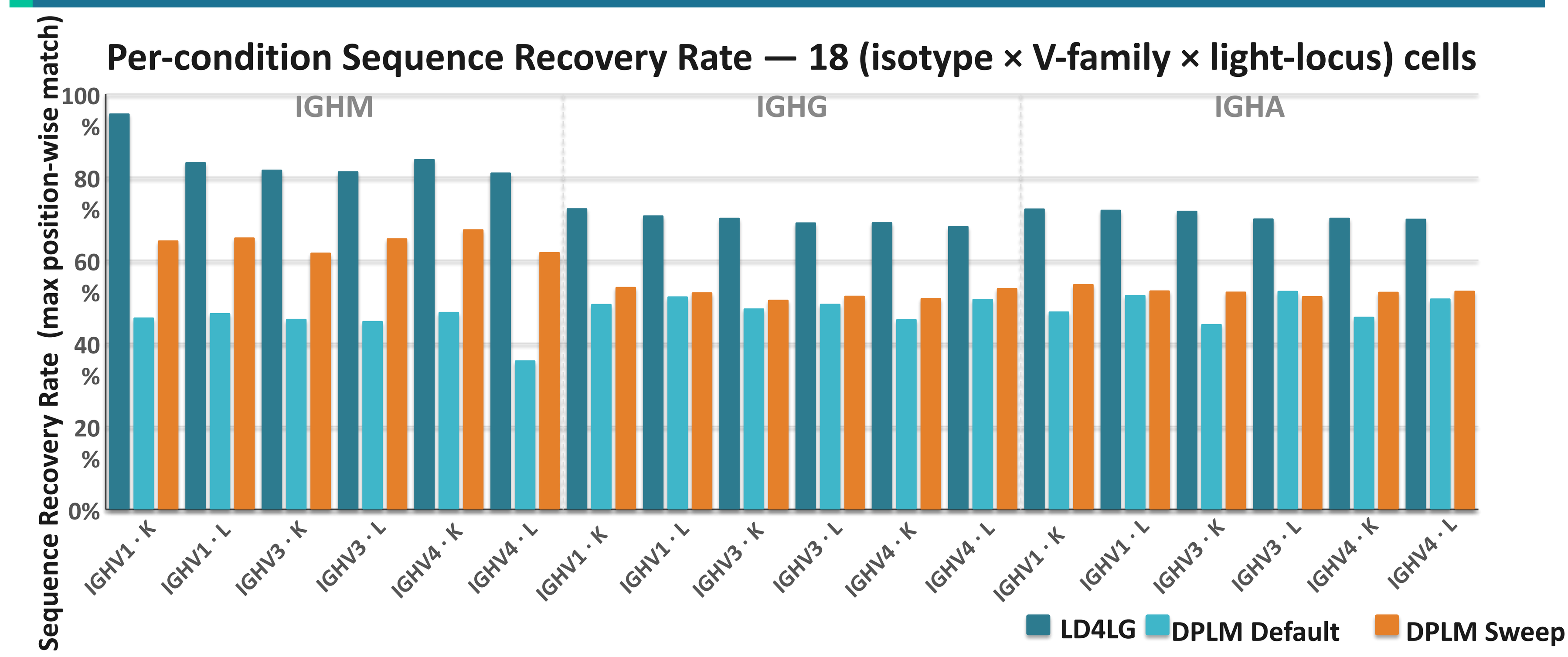



*97% → 26% : 71-pp drop in foldable share traded for 4× more diversity*

## §4.3 Quality–Diversity Pareto




## §4.4 Per-condition Recovery




## §5 Conclusion

✓ **Decoding controls the quality–diversity trade-off**
✓ **DPLM is stable but lacks diversity**
✓ **CFG enables controllable generation**

## §6 Challenges + Future

### Challenges:

- Balancing diversity and structural quality remains difficult
- Distributed training instability (DDP divergence) ⟶ limited single-GPU runs

### Future directions:

- Better decoding strategies to improve the quality–diversity trade-off
- Hybrid approaches combining latent and token-space diffusion
- Extend to paired TCR (β + α) and cross species

★ EXTENSIONS BEYOND PAPER

## §7 References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NSF ACCESS — GPU compute via Purdue Anvil. allocations.access-ci.org
Karpathy. autoresearch. github.com/karpathy/autoresearch
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